06/09/2026

Code-F_Phase-01

Aakash

Medical | IIT-JEE | Foundations

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MM : 720 | **Term Exam for NEET-2027 (XII Studying) Ph-1 - TE03F** | **Time : 180 Min.**

Topics Covered:
Physics: Electromagnetic Induction, Alternating Current, Electromagnetic Waves
Chemistry: Coordination Compounds, Haloalkanes and Haloarenes
Botany: Molecular Basis of Inheritance
Zoology: Evolution

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

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$\vec{E} = E_{01}\cos\left[2\pi f\left(\frac{z}{c} - t\right)\right]\hat{x}$

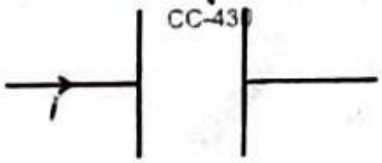
PHYSICS

1. The amplitude of the magnetic field of an electromagnetic wave in vacuum is $B_0 = 510$ nT. The amplitude of the electric field in the wave is
(1) 120 N/C (2) 134 N/C
(3) 510 N/C (4) 153 N/C

2. The electromagnetic radiation which has the largest wavelength among the following is
(1) x-rays (2) Gamma rays
(3) Ultraviolet rays (4) Microwaves

3. If $\vec{E}$ and $\vec{B}$ represent electric and magnetic field vectors of an *EM* wave. The direction of propagation wave is along
(1) $\vec{E}$ (2) $\vec{B}$
(3) $\vec{E} \times \vec{B}$ (4) $\vec{B} \times \vec{E}$

4. A parallel plate capacitor with circular plates of radius R is being charged with a current I as shown in the figure. At the instant shown, the displacement current between the plates enclosed between $\frac{R}{4}$ and R is given by



(1) $\frac{3I}{4}$ (2) $\frac{15I}{16}$
(3) I (4) $\frac{I}{4}$

5. In an electromagnetic wave, the amplitude of electric field is 2 V/m. The average energy density of wave is
(1) 17.7×10^{-12} J/m^3 (2) 8.85×10^{-12} J/m^3
(3) 4.42×10^{-12} J/m^3 (4) 1×10^{-12} J/m^3

6. An *EM* wave from air enters in a medium. The electric fields are $\vec{E}_1 = E_{01}\cos\left[2\pi f\left(\frac{z}{C} - t\right)\right]\hat{x}$ in air and $\vec{E}_2 = E_{02}\cos[K(3z - ct)]\hat{x}$ in medium where the wave number K and frequency f refer to their values in air. The medium is non-magnetic. If ϵ_{r_1} and ϵ_{r_2} refer to relative permittivity's of air and medium respectively, then which of the following option is correct?
(1) $\frac{\epsilon_{r_1}}{\epsilon_{r_2}} = 9$ (2) $\frac{\epsilon_{r_1}}{\epsilon_{r_2}} = \frac{1}{9}$
(3) $\frac{\epsilon_{r_1}}{\epsilon_{r_2}} = 4$ (4) $\frac{\epsilon_{r_1}}{\epsilon_{r_2}} = \frac{1}{4}$

7. A parallel plate capacitor is charged by connecting it to a battery through a resistor. If I is the current in the circuit, then in the gap between the plates:
(1) There is no current
(2) Displacement current of magnitude equal to I flows in the same direction as I
(3) Displacement current of magnitude equal to I flows in a direction opposite to that of I
(4) Displacement current of magnitude greater than I flows but can be in any direction

8. Match List-I with List-II, where symbols have their usual meaning.

List-I	List-II
a. Gauss's law in electrostatics	(i) $\oint \vec{E}.\vec{dl} = -\frac{d\phi_B}{dt}$
b. Faraday's law	(ii) $\oint \vec{B}.d\vec{s} = 0$
c. Gauss's law in magnetism	(iii) $\oint \vec{B}.\vec{dl} = \mu_0 i_c + \mu_0\varepsilon_0\frac{d\phi_E}{dt}$
d. Ampere-Maxwell Law	(iv) $\oint \vec{E}.\vec{ds} = \frac{q}{\varepsilon_0}$

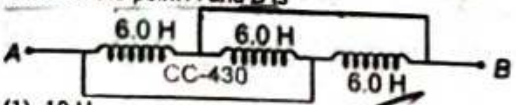
Choose the correct answer from the options given below.
(1) a(ii), b(iii), c(iv), d(i) (2) a(iii), b(iv), c(i), d(ii)
(3) a(iv), b(i), c(ii), d(iii) (4) a(i), b(ii), c(iii), d(iv)

9. The maximum radiation pressure exerted by sunlight in space (I = 1350 W/m^2) on a highly polished silver surface?
(1) 1.4×10^{-2} Pa (2) 0.12 Pa
(3) 9×10^{-6} Pa (4) 9.5×10^{-5} Pa

10. The magnetic field in a plane electromagnetic wave is given by $B_y = 2 \times 10^{-7}\sin(0.5 \times 10^3 x + 1.5 \times 10^{11} t)$ T where x, t are in m and second respectively. The expression for the electric field is
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(1) $E_z = 60\sin(0.5 \times 10^3 y + 1.5 \times 10^{11} t)$ V/m
(2) $E_z = 2 \times 10^{-7}\sin(0.5 \times 10^3 x + 1.5 \times 10^{11} t)$ V/m
(3) $E_z = 60\sin(0.5 \times 10^3 x + 1.5 \times 10^{11} t)$ V/m
(4) $E_z = 2 \times 10^{-7}\sin(0.5 \times 10^3 y + 1.5 \times 10^{11} t)$ V/m

11. If ε_0 and μ_0 are the permittivity and permeability of vacuum, while ε and μ are the permittivity and permeability for a medium, then the refractive index of the medium is given by
(1) $\sqrt{\mu_0\varepsilon_0}$ (2) $\sqrt{\mu\varepsilon}$
(3) $\sqrt{\frac{\mu\varepsilon}{\mu_0\varepsilon_0}}$ (4) $\sqrt{\frac{\mu_0\varepsilon_0}{\mu\varepsilon}}$

12. The rate of increase of current through a inductor is 2.4 A/s. If voltage drop across the inductor is 1.6 V then self inductance of the inductor will be
(1) $\frac{2}{3}$ H
(2) $\frac{1}{3}$ H
(3) $\frac{1}{2}$ H
(4) $\frac{1}{3}$ H

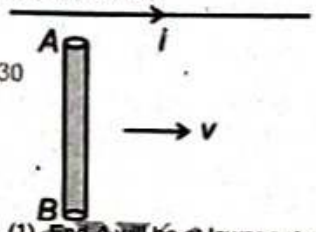
13. Three inductor each of inductance of 6.0 H is connected as shown below. The equivalent inductance of the circuit between the point A and B is


(1) 18 H (2) 2 H
(3) 6 H (4) Zero

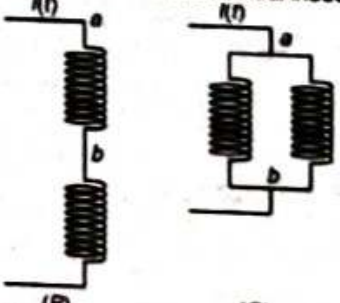
14. In the given diagram, a line of force of a particular force field is shown. Out of the following options, it can represent

(1) An electrostatic field only
(2) A gravitational field of a mass at rest only
(3) Both (1) & (2)
(4) An induced electric field

15. The magnitude flux ϕ (in Weber) in a closed circuit of resistance 10 Ω varies with time t (in second) according to the equation $\phi = 6t^2 - 5t + 1$. The magnitude of induced current at t = 0.25 s is
(1) 0.2 A (2) 2 A
(3) 1 A (4) 3 A

16. The current carrying wire and the rod AB are in the same plane. The rod moves parallel to the wire with a velocity v. Which one of the following statements is true about induced emf in the rod?
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(1) End A will be at lower potential with respect to B
(2) A and B will be at the same potential
(3) There will be no induced emf in the rod
(4) Potential at A will be higher than that at B

17. Two identical inductors are connected in two different configurations P and Q, where a time varying current $I(t)$ is flowing, as shown in the figure. The induced emf between points a and b for configuration P is E_P and that for configuration Q is E_Q. The ratio E_P/E_Q is:
[Neglect the effect of mutual inductance.]


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(1) $\frac{1}{4}$ (2) 1
(3) 2 (4) $\frac{1}{2}$

A thin semicircular co... with its plane ve... shown in fi... the ring ...

$\oint \vec{E}.\vec{dl} = -\frac{d\phi_B}{dt}$ | $\oint \vec{B}.\vec{ds} = 0$ | $\oint \vec{B}.\vec{dl} = \mu_0 i_c + \mu_0\varepsilon_0\frac{d\phi_E}{dt}$ | $\oint \vec{E}.\vec{ds} = q/\varepsilon_0$

0.063

inductor = 2.4 A/s
drop voltage → 1.6
$\frac{24}{10} \div \frac{16}{10} = \frac{4}{1}$

$(t = 0.25)$
$\phi = 6t^2 - 5t + 1$
$6(0.25)^2 - 5(0.25) + 1$
$\Rightarrow 6 \times 0.5 - 5(0.25) + 1$
$\Rightarrow 3 - 1.25 + 1$
$\Rightarrow 2.75$

ε_0 μ_0 → $\sqrt{\frac{\mu\varepsilon}{\mu_0\varepsilon_0}}$

$I = 1350$ W/m^2

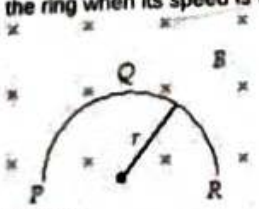
$B_y = 2 \times 10^{-7}\sin(0.5 \times 10^3 x + 1.5 \times 10^{11} t)$ T

(x, t) = m & second

$E_z = 2 \times 10^{-7}\sin(0.5 \times 10^3 y + 1.5 \times 10^{11} t)$

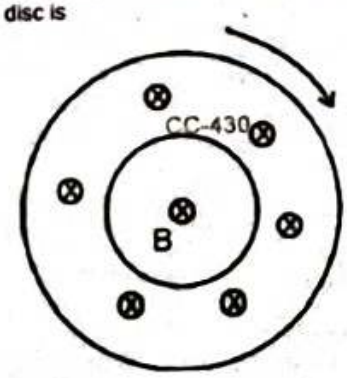
$E_x = 60\sin(0.5 \times 10^3 y + 1.5 \times 10^{11} t)$ V/m
$E_x = 2 \times 10^{-7}\sin(0.5 \times 10^3 x + 1.5 \times 10^{11} t)$ V/m
$E_z = 60\sin(0.5 \times 10^3 x + 1.5 \times 10^{11} t)$ V/m

A thin semicircular conducting ring (PQR) of radius r is falling with its plane vertical in a horizontal magnetic field B, as shown in figure. The potential difference developed across the ring when its speed is v, is



(1) Zero
(2) $Bv\pi r^2/2$ and P is at higher potential
(3) πrBv and R is at higher potential
(4) $2rBv$ and R is at higher potential

19. An annular disc of copper with inner and outer radius R and 2R is rotating with a uniform angular speed ω in a region where a uniform magnetic field B exist as shown. The net emf induced between the inner edge and outer edge of the disc is



(1) $\frac{B\omega R^2}{2}$
(2) $\frac{3B\omega R^2}{2}$
(3) $2B\omega R^2$
(4) Zero

20. A small square loop of side l is placed inside a large square loop of side L ($L \geq l$). The loops are coplanar and their centres coincide. The mutual inductance of the system is

(1) $\frac{2\mu_0 l^2}{\pi L}$
(2) $\frac{2\sqrt{2}\mu_0 l^2}{\pi L}$
(3) $\frac{2\mu_0 L^2}{\pi l}$
(4) $\frac{2\sqrt{2}\mu_0 L^2}{\pi l}$

21. A long solenoid of diameter 0.1 m has 2×10^4 turns per meter. At the centre of the solenoid, a coil of 100 turns and radius 0.01 m is placed with its axis coinciding with the solenoid axis. The current in the solenoid reduces at a constant rate to 0 A from 4 A in 0.05 s. If the resistance of the coil is $10\pi^2\ \Omega$, the total charge flowing through the coil during this time is

(1) 32 π μC
(2) 16 μC
(3) 32 μC
(4) 16π μC

22. Faradays law of electromagnetic induction are consequences of conservation of

(1) Energy
(2) Force
(3) Momentum
(4) Magnetic field

23. A 10 mH coil carries a current of 1 A. Energy stored in its magnetic field is

(1) 1 mJ
(2) 2 mJ
(3) 3 mJ
(4) 5 mJ

24. For AC circuit capacitive reactance and Time period of AC are plotted. The nature of graph is

(1) Straight line with negative slope
(2) Parabola
(3) Hyperbola
(4) Straight line with +ve constant slope

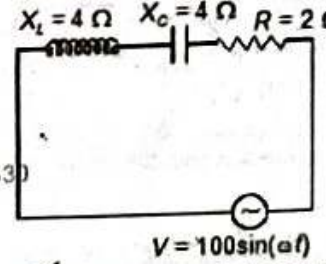
25. A transformer has 50 turns in primary and 200 turns in secondary. If 8 V DC source is connected across primary, the output voltage is

(1) 32 V
(2) 16 V
(3) 8 V
(4) Zero

26. A resistor and a capacitor are connected in series with an AC source. If voltage across the resistor is 40 V and voltage across the capacitor is 30 V then voltage of the source is

(1) 30 V
(2) 40 V
(3) 50 V
(4) 10 V

27. Which of the following statement is correct regarding the AC circuit shown in the adjacent figure?

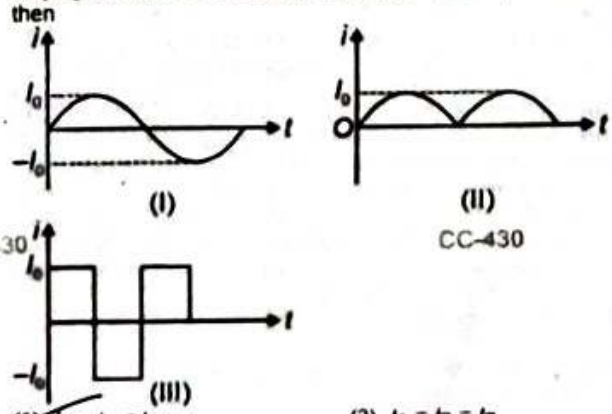


(1) Average power dissipated in the circuit is 2500 W
(2) The phase difference between source emf and current is $\phi = \cos^{-1}\left(\frac{2}{5}\right)$
(3) The rms value of current through the circuit is $I_{rms} = 5\sqrt{2}$ A
(4) All of these

28. The inductive reactance and resistance of L-R circuit are respectively 30 ohm and 40 ohm. The impedance of the circuit is

(1) 50 ohm
(2) 10 ohm
(3) 70 ohm
(4) $\frac{1}{3}$ ohm

29. If I_1, I_2 and I_3 are the respective r.m.s. values of the time varying currents as shown in the three cases I, II and III, then



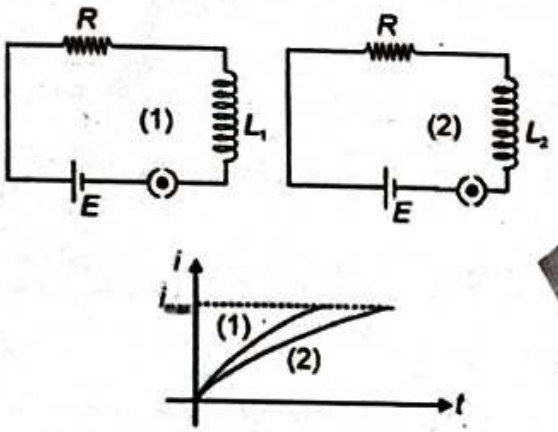
(1) $I_1 = I_2 < I_3$
(2) $I_1 = I_2 = I_3$
(3) $I_1 > I_2 = I_3$
(4) $I_1 \neq I_2 \neq I_3$

30. An ac circuit consists of an inductor of inductance 0.5 H and a capacitor of capacitance 8 μF in series. The current in the circuit is maximum when the angular frequency of ac source is
(1) 500 rad/s (2) 200 rad/s
(3) 300 rad/s (4) 400 rad/s

31. The voltage of an source varies with time according to the equation, $V = 100 \sin 100 \pi t \cos 100 \pi t$. Where t is in second and V is in volt, then
(1) The peak voltage of the source is 100 volt
(2) The peak voltage of source is 50 volt
(3) The peak voltage of source is $100/\sqrt{2}$ volt
(4) The peak voltage of source is $100\sqrt{2}$ volt

32. An AC source of angular frequency ω is fed across a resistance R and a capacitor C in series. The current registered is I. If now the frequency of source is changed to $\frac{\omega}{3}$ (but supply voltage remaining same), then the current in the circuit is found to be halved. The ratio of reactance to resistance at the original frequency ω is
(1) $\sqrt{\frac{2}{5}}$ (2) $\sqrt{\frac{3}{5}}$
(3) $\sqrt{\frac{1}{3}}$ (4) $\sqrt{\frac{5}{4}}$

33. There are two CR circuits as shown. Key is closed simultaneously at $t = 0$. Current growth in both circuits are as shown, in figure

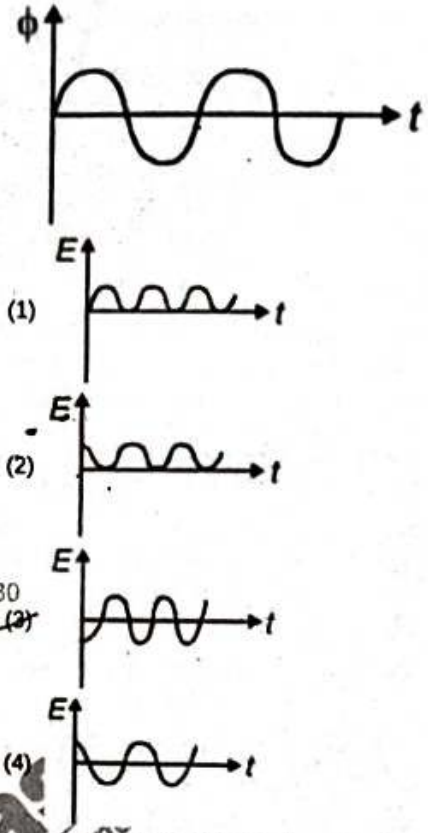


Then
(1) $L_1 > L_2$ (2) $L_1 < L_2$
(3) $L_1 = L_2$ (4) $L_1 \geq L_2$

34. A transformer having efficiency of 80% is working on 2 kW power supply. If the current in the secondary coil is 3.2 A, the voltage across the secondary coil will be
(1) 1000 V (2) 400 V
(3) 700 V (4) 500 V

35. **Statement A:** In an AC circuit containing pure inductance, higher the frequency of A.C., higher is the inductive reactance offered by inductor in A.C. circuit.
Statement B: Inductor offers no opposition to the flow of d.c current.
Consider the given statements and choose the correct option
(1) Both statements are true
(2) Both statements are false
(3) Statement A is true but B is false
(4) Statement B is true but A is false

36. The magnetic flux through a coil varies with time t as shown in the diagram. Which graph best represents the variation of the e.m.f. E induced in the coil with time t?



37. Eddy currents are induced when
(1) A metal block is kept in a changing magnetic field
(2) A metal block is kept in a uniform magnetic field
(3) A coil is kept in a uniform magnetic field
(4) Current is passed in a coil

38. The magnetic energy stored in a long solenoid of area of cross-section A in a small region of length L is
(1) $\frac{B^2AL}{2\mu_0^2}$ (2) $\frac{AL}{2\mu_0}$
(3) $\frac{1}{2}\mu_0 B^2 AL$ (4) $\frac{B^2AL}{2\mu_0}$

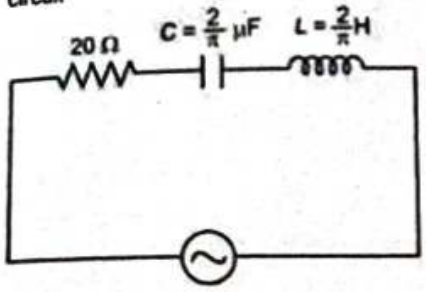
39. The mean value of current $I = I_0 \cos\omega t$ from $t = 0$ to $t = \frac{\pi}{\omega}$ is
(1) Zero (2) $\frac{I_0}{\pi}$
(3) $\frac{2I_0}{\pi}$ (4) $\frac{I_0}{2\pi}$

40. A sinusoidal supply of frequency 10 Hz and r.m.s. voltage 12 V is connected to a 2.1 μF capacitor. What is r.m.s. value of current?
(1) 5.5 mA (2) 20 mA
(3) 26 mA (4) 1.6 mA

41. Power factor of an ideal choke coil (i.e., $R = 0$) is
(1) Near about zero (2) Zero
(3) Near about one (4) One

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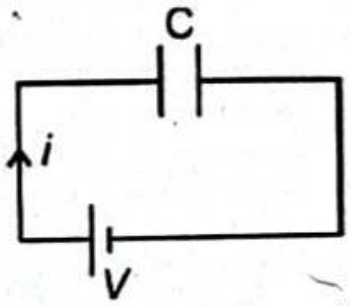
42. In an AC circuit as shown in the figure, the source is of r.m.s voltage 200 V and variable frequency. At resonance, the circuit



(1) Has a r.m.s current 10 A
(2) Has a resonant frequency 250 Hz
(3) Has voltage across capacitor, inductor and resistor in same phase
(4) Both (1) and (2)

43. EM waves can transport
(1) Momentum
(2) Energy
(3) Charge
(4) Both (1) & (2)

44. At a particular instant the current in the circuit given below is i. The displacement current between the plates of the capacitor shown below is



(1) Zero
(2) i
(3) $\frac{i}{2}$
(4) $\frac{i}{4}$

45. If electric field $E = E_0 \sin(\omega t - kz)\hat{i}$ and magnetic field $B = B_0 \sin(\omega t - kz)\hat{j}$, then electromagnetic wave travels along
(1) $+z$ direction
(2) $-y$ direction
(3) $+x$ direction
(4) $+y$ direction

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CHEMISTRY

46. Match list-I with list-II

	List-I (Octahedral complex)		List-II (Moles of AgCl precipitated per mole of the compound with excess $AgNO_3$)
a.	$CoCl_3 \cdot 4NH_3$	(i)	0
b.	$CoCl_3 \cdot 5NH_3$	(ii)	3
c.	$CoCl_3 \cdot 3NH_3$	(iii)	2
d.	$CoCl_3 \cdot 6NH_3$	(iv)	1

The correct match is
(1) a(iv), b(iii), c(i), d(ii)
(2) a(i), b(iii), c(ii), d(iv)
(3) a(ii), b(i), c(iii), d(iv)
(4) a(iii), b(ii), c(iv), d(i)

47. The sum of coordination number and oxidation number of the metal M in the complex
$[M(en)_2(C_2O_4)]Cl$
(Where en is ethylenediamine) is
(1) 7
(2) 8
(3) 9
(4) 6

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48. The complex which has *facial* and *meridional* isomers is
(Given : py = pyridine and en = $H_2N - CH_2 - CH_2 - NH_2$)
(1) $[Cr(py)_3(Cl)_3]$
(2) $[Cr(H_2O)_6]^{3+}$
(3) $[Co(NH_3)_4(H_2O)_2]^{3+}$
(4) $[Ni(en)_2(H_2O)_2]^{2+}$

49. According to crystal field theory, the correct order of ligands with respect to their decreasing order of field strength is
(1) $CO > H_2O > NH_3 > Cl^-$
(2) $Cl^- > H_2O > NH_3 > CO$
(3) $Cl^- > NH_3 > H_2O > CO$
(4) $CO > NH_3 > H_2O > Cl^-$

50. Dimethyl glyoxime reagent is commonly used as co-ordinating reagent in the qualitative estimation of
(1) Pb^{2+}
(2) Mg^{2+}
(3) Ag^+
(4) Ni^{2+}

51. Vitamin B_{12} is a coordination compound of
(1) Copper
(2) Iron
(3) Magnesium
(4) Cobalt

52. The inner orbital complex among the following is
(1) $[MnCl_6]^{3-}$
(2) $[Fe(H_2O)_6]^{3+}$
(3) $[Co(NH_3)_6]^{3+}$
(4) $[MnF_6]^{3-}$

53. What is the correct electronic configuration of the central atom in $K_2[NiCl_4]$ based on crystal field theory?
(1) $t_{2g}^6 e_g^2$
(2) $t_{2g}^6 e_g^0$
(3) $e^4 t_2^2$
(4) $e^4 t_2^4$

54. Formula of the compound tetraammineaquachloridocobalt(III) chloride is
(1) $[Co(NH_3)_4(H_2O)Cl]Cl_2$
(2) $[Co(NH_3)_4Cl_2]Cl.H_2O$
(3) $[Co(NH_3)_3Cl_3].H_2O$
(4) $[Co(NH_3)_3Cl(H_2O)]Cl_2$

55. The compound which will show optical isomerism is
(1) trans-$[CrCl_2(ox)_2]^{3-}$
(2) $[Co(en)_3]^{3+}$
(3) $[Co(NH_3)(NO_2)_3]$
(4) $[Co(NH_3)_5Cl]^{2+}$

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56. Which of the following carbonyls will have the strongest C – O bond?
(1) $[Mn(CO)_6]^+$
(2) $[V(CO)_6]^-$
(3) $[Fe(CO)_5]$
(4) $[Cr(CO)_6]$

57. Wilkinson catalyst is
(1) $[Co(Ph_3P)_3Cl]$
(2) $K[Au(CN)_2]$
(3) $[Rh(Ph_3P)_3Cl]$
(4) $Ni(CO)_4$

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58. Which of the following complex can show linkage as well as ionisation isomerism?
(1) $[Cr(H_2O)_6]Cl_3$ (2) $[Cr(H_2O)_3(CN)_3]Cl_3$
(3) $[Cr(NH_3)_3Br_3]Cl_3$ (4) $[Cr(H_2O)_4Cl_2]Cl$

59. Match List I with List II.

	List I (Complex)		List II (Type of isomerism)
A.	$[Co(NH_3)_5(NO_2)]Cl_2$	I.	Solvate isomerism
B.	$[Co(NH_3)_5(SO_4)]Br$	II.	Linkage isomerism
C.	$[Co(NH_3)_6][Cr(CN)_6]$	III.	Ionization isomerism
D.	$[Co(H_2O)_6]Cl_3$	IV.	Coordination isomerism

Choose the **correct** answer from the options given below:
(1) A-II, B-III, C-IV, D-I (2) A-I, B-III, C-IV, D-II
(3) A-I, B-IV, C-III, D-II (4) A-II, B-IV, C-III, D-I

60. Match List-I with List-II.

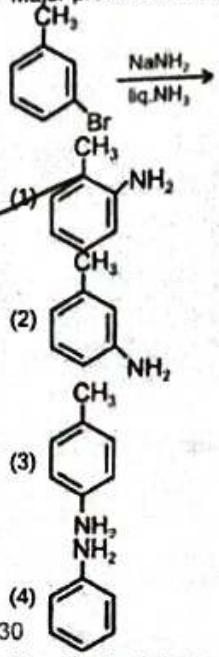
	List-I (Co-ordination complex)		List-II (Magnetic moment in B.M)
a.	$[Fe(CN)_6]^{3-}$	(i)	5.92 BM
b.	$[Fe(H_2O)_6]^{3+}$	(ii)	0 BM
c.	$[Fe(CN)_6]^{4-}$	(iii)	4.90 BM
d.	$[Fe(H_2O)_6]^{2+}$	(iv)	1.73 BM

Choose the correct answer from the options given below.
(1) a(iv), b(i), c(ii), d(iii) (2) a(iv), b(ii), c(i), d(iii)
(3) a(ii), b(iv), c(iii), d(i) (4) a(i), b(iii), c(iv), d(ii)

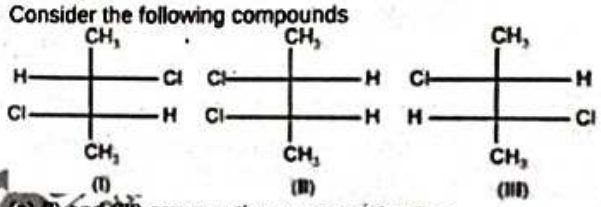
61. The correct order of the wavelength of light absorbed by the following complexes is,
A. $[Co(NH_3)_6]^{3+}$
B. $[Co(CN)_6]^{3-}$
C. $[Cu(H_2O)_4]^{2+}$
D. $[Ti(H_2O)_6]^{3+}$
Choose the correct answer from the options given below:
(1) C < D < A < B (2) C < A < D < B
(3) B < D < A < C (4) B < A < D < C

62. Given below are two statements,
Statement I: $[Co(NH_3)_3(NO_2)_3]$ is heteroleptic complex while $[Co(C_2O_4)_3]^{3-}$ is homoleptic complex
Statement II: Both $[Co(NH_3)_3(NO_2)_3]$ and $[Co(C_2O_4)_3]^{3-}$ exhibit optical isomerism.
In the light of above statements, choose the correct option.
(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

63. Major product of the given reaction is



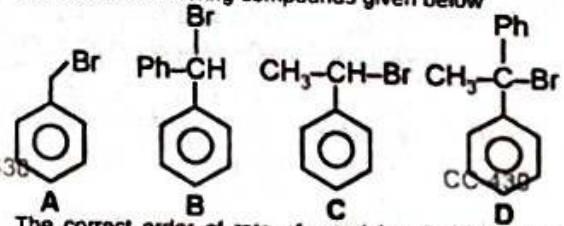
64. Consider the following compounds



(a) (I) and (III) are enantiomers to each other
(b) (I) and (II) are diastereomers to each other
(c) (II) and (III) are diastereomers to each other
The correct statement(s) is/are
(1) (c) only (2) (b) and (c) only
(3) (a) and (b) only (4) (a), (b) and (c)

65. $CH_3-C(CH_3)=CH_2 + HBr \rightarrow$ X (major product) $\xrightarrow{\text{hydrolysis}}$ Y
Compound Y is
(1) An alkane (2) An ether
(3) An aldehyde (4) An alcohol

66. Consider the following compounds given below

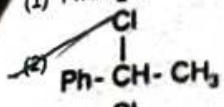


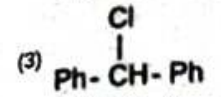
The correct order of rate of reactivity of given compounds towards S_N1 reaction is
(1) A > B > C > D (2) D > A > C > B
(3) D > B > C > A (4) C > B > D > A

67. Compound which shows optical activity is
(1) Ethanol (2) Butan-2-ol
(3) Butan-1-ol (4) Propan-2-ol

Which of the following compounds will undergo racemisation on hydrolysis with NaOH?

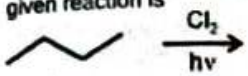
(1) $PhCH_2Cl$

(2) 

(3) 

(4) $(CH_3)_3C - Cl$

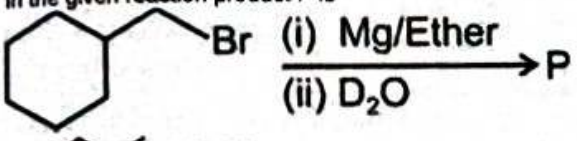
69. Total number of monochlorinated products formed for the given reaction is

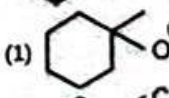


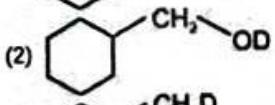
(1) 3 (2) 5

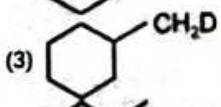
(3) 2 (4) 4

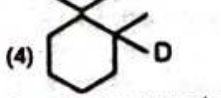
70. In the given reaction product P is



(1) 

(2) 

(3) 

(4) 

71. Consider the following statements.

Statement (I) : Haloarenes are less reactive than haloalkanes towards nucleophilic substitution reaction.

Statement (II) : In haloarenes, the electron pairs on halogen atom are in conjugation with π-electrons of the ring.

In the light of above statements, choose the correct option.

(1) Statement (I) is correct but Statement (II) is incorrect

(2) Statement (I) is incorrect but Statement (II) is correct

(3) Both Statement (I) and Statement (II) are correct

(4) Both Statement (I) and Statement (II) are incorrect

72. Given below are two statements.

Statement I : Meso compounds are optically inactive.

Statement II : Meso compounds does not have any chiral carbon.

In the light of above statements, choose the correct option.

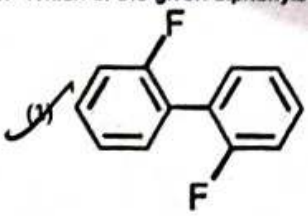
(1) Statement I is correct but statement II is incorrect

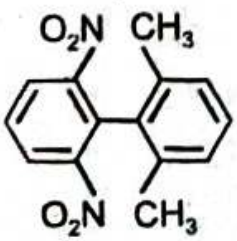
(2) Statement I is incorrect but statement II is correct

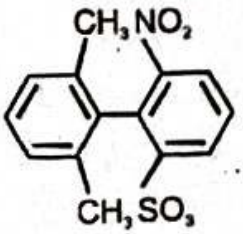
(3) Both statements I and II are correct

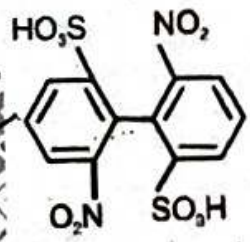
(4) Both statements I and II are incorrect

73. Which of the given biphenyls can be optically active?

(1) 

(2) 

(3) 

(4) 

74. Given below are two statements.

Statement I : Chloroform is stored in the dark coloured bottles and filled till the top.

Statement II : Chloroform gets oxidised to poisonous phosgene gas.

In the light of above statements, choose the correct option.

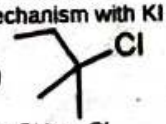
(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

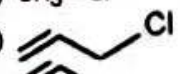
(3) Both statements I and II are correct

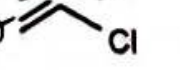
(4) Both statements I and II are incorrect

75. The compound which will react at fastest rate by S_N2 mechanism with KI is

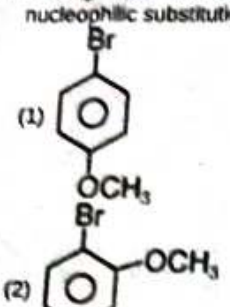
(1) 

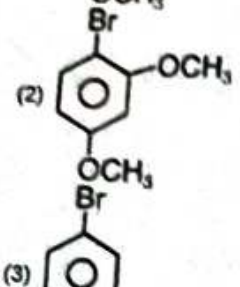
(2) $CH_3 - Cl$

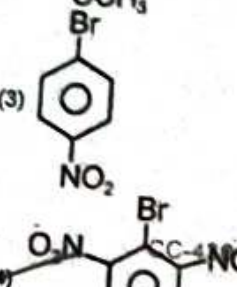
(3) 

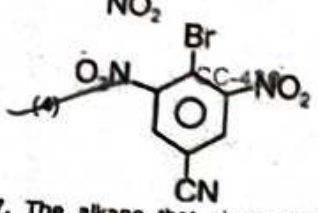
(4) 

76. Among the following, which compound undergoes nucleophilic substitution reaction most easily?

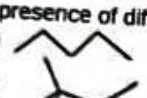
(1) 

(2) 

(3) 

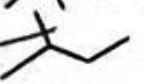
(4) 

77. The alkane that gives maximum number of mono-chloro products on chlorination (excluding stereoisomers) with Cl_2 in presence of diffused sunlight is

(1) 

(2)

(3) 

(4) 

78. Consider the following statements:

Statement I : A mixture of an alkyl halide and aryl halide gives an alkylarene when treated with sodium in dry ether is called Fittig reaction.

Statement II: For the preparation of 2, 3-Dimethylbutane by Wurtz reaction, the alkyl halide needed as substrate is isopropyl chloride.

In the light of above statements, choose the **correct** option.

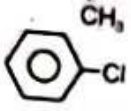
(1) Statement I is incorrect but Statement II is correct
(2) Both statement I and statement II are correct
(3) Both statement I and statement II are incorrect
(4) Statement I is correct but Statement II is incorrect

79. The compound which is most difficult to hydrolyse is

(1) $CH_2 = CH - CH_2Cl$

(2) $(CH_3)_3C - Cl$

(3) $CH_3 - \underset{\displaystyle CH_3}{\overset{\displaystyle |}{CH}} - Cl$

(4) 

80. Given below are the two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Haloalkanes react with AgCN to form alkyl cyanide as the major product.

Reason (R) : AgCN is mainly covalent in nature and nitrogen is free to donate electron pair forming cyanide as the major product.

In the light of above statement, choose the **correct** option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

81. Which of the following has primary valency +2?

(1) $[Co(NH_3)_5Cl]Cl$ (2) $[Co(NH_3)_4Cl_2]Cl$
(3) $[Co(NH_3)_3Cl_3]$ (4) $[Co(H_2O)_4Cl_2]Cl$

82. Which of the following will have greatest molar conductance at infinite dilution?

(1) $[Pt(NH_3)_5Cl]Cl_3$ (2) $[Pt(NH_3)_4Cl_2]Cl_2$
(3) $[Pt(NH_3)_2Cl_4]$ (4) $[Pt(NH_3)_6]Cl_4$

83. How many hydrogen bonded water molecules are present in blue vitriol?

(1) Zero (2) One
(3) Four (4) Five

84. The complex compound having square planar geometry is

(1) $[Ni(CO)_4]$ (2) $[Ni(CN)_4]^{2-}$
(3) $[Mn(CN)_6]^{3-}$ (4) $[MnCl_4]^{2-}$

85. The complex compound having maximum magnetic moment is

(1) $[CoF_6]^{3-}$ (2) $[Cr(NH_3)_6]^{3+}$
(3) $[FeF_6]^{3-}$ (4) $[Mn(CN)_6]^{4-}$

86. Which one of the following is gem-dihalide?

(1) CH_3CHBr_2 (2) $CH_2Br - CH_2Br$
(3) $CH_3CHBrCH_2Br$ (4) $CH_3CHBrCH_2CH_2Br$

87. IUPAC name of neopentyl bromide is

(1) 1-Bromo-4, 4-dimethylpentane
(2) 1-Bromo-3-methylbutane
(3) 1-Bromo-2, 2-dimethylpropane
(4) 2-Bromo-2-methylbutane

88. Which among the given compounds has the highest dipole moment?

(1) CH_3F (2) CH_3Br
(3) CH_3I (4) CH_3Cl

89. Swart's reaction is used for the preparation of

(1) $CH_3 - Cl$ (2) $CH_3 - Br$
(3) $CH_3 - I$ (4) $CH_3 - F$

90. Which among the following is Darzen's method of preparation of alkyl chloride?

(1) $CH_3CH_2OH \xrightarrow{PCl_5} CH_3CH_2Cl$

(2) $CH_3CH_2OH \xrightarrow[ZnCl_2]{HCl} CH_3CH_2Cl$

(3) $CH_3CH_2OH \xrightarrow{PCl_3} CH_3CH_2Cl$

(4) $CH_3CH_2OH \xrightarrow{SOCl_2} CH_3CH_2Cl$

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BOTANY

91. Select the **incorrect** statement w.r.t. polymorphism.
(1) Polymorphism arises due to mutations
(2) Polymorphism in DNA sequence is the basis of DNA fingerprinting
(3) Polymorphism represents variation at genetic level
(4) Polymorphism do not play any role in evolution and speciation

92. Read the following Assertion and Reason and select the **correct** option.
Assertion : It is considered that RNA was the first genetic material.
Reason : RNA has evolved from DNA with chemical modifications that make it capable to carry information from one generation to the next.
(1) Both Assertion and Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion and Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

93. What will happen if the DNA isolated from virulent *Streptococcus* bacteria is kept in a test tube where non-virulent *Streptococcus* bacteria are growing?
(1) Isolated DNA will be replaced by RNA
(2) Isolated DNA will be rapidly degraded into nucleotides .
(3) Isolated DNA will form separate chromosome in non-virulent form
(4) There may be incorporation of DNA in genome of non-virulent bacteria

94. Match the column I with column II

Column I	Column II
a. RNA polymerase II	(i) Release factor
b. Initiation of transcription	(ii) tRNA
c. Termination of translation	(iii) σ factor
d. Clover leaf like structure	(iv) hnRNA

Choose the **correct** answer from the options given below.
(1) a(ii), b(iv), c(i), d(iii)
(2) a(iii), b(ii), c(iv), d(i)
(3) a(iii), b(iv), c(i), d(ii)
(4) a(iv), b(iii), c(i), d(ii)

95. The contribution of Marshall Nirenberg is
(1) Development of cell-free system for protein synthesis.
(2) Development of chemical method for synthesising RNA molecules with defined combinations of bases.
(3) The development of the actual structure of the tRNA
(4) The proposition that the codon is read in mRNA in a contiguous fashion.

96. EST refers to:
(1) Introduction of vector DNA into the host
(2) Identifying all the genes that are expressed as RNA
(3) Finding chromosomal locations for disease-associated sequences.
(4) Blind approach of sequencing the whole genome and later assigning the function to different regions

97. In Meselson and Stahl experiment, density gradient was formed by using
(1) CsCl
(2) ^{3}H-thymidine
(3) Sucrose
(4) Glucose

98. For the codon 5' UAC 3', what could be the sequence of anticodon on tRNA?
(1) 5' ATG 3'
(2) 5' GTA 3'
(3) 5' AUG 3'
(4) 5' GUA 3'

99. Read the following statements and state them as true **(T)** or false **(F)**.
(A) Tailing and synthesis of primary transcript are coupled in prokaryotes.
(B) A eukaryotic cell has more than three types of RNA polymerases.
(C) Splicing represents the dominance of RNA world.
(D) In tailing, adenylate residue are added in template dependent manner.
Choose the appropriate option.

	A	B	C	D
(1)	F	T	T	F
(2)	F	T	T	T
(3)	F	F	F	T
(4)	T	F	T	F

(1) (1)
(2) (2)
(3) (3)
(4) (4)

100. Match List - I with List - II.

	List - I		List - II
a.	Hershey and Chase	i	Conducted the transformation experiment in-vitro.
b.	Taylor *et al*	ii	Worked on *Streptococcus* and mice.
c.	Frederick Griffith	iii	Used radioisotopes to trace the mechanism of DNA replication in a photoautotroph.
d.	Avery, MacLeod and McCarty	iv	Provided unequivocal proof of DNA acting as genetic material.

Choose the most appropriate option.
(1) a-ii , b-iii , c-iv , d-i
(2) a-iv , b-iii , c-ii , d-i
(3) a-ii , b-iii , c-i , d-iv
(4) a-iv , b-i , c-ii , d-iii

101. How many phosphodiester bonds are present in genetic material that is a linear dsDNA molecule?
(1) It is equivalent to the number of 3' hydroxyl group present in that DNA
(2) It is equivalent to the number of nucleotides present in one strand of DNA
(3) It is two less than the number of nucleotides present in that dsDNA
(4) It is one less than the total number of nucleotides present in that dsDNA

102. The structural gene in *lac* operon is responsible for all, **except**
(1) Synthesis of product that binds to the operator region when inducer is absent
(2) Formation of product that increases the permeability of the cell to β-galactosides
(3) Synthesis of enzyme that hydrolyses the lactose
(4) Synthesis of enzyme that transfers acetyl group to β-galactoside

103. 'UGA does not code for any amino acid'. All of the below given conclusions can be withdrawn from the above statement, **except**
(1) UGA is a nonsense codon
(2) There is no tRNA for the codon UGA
(3) Genetic code is degenerate
(4) Translation terminates when ribosome reaches UGA of mRNA

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104. During DNA replication SSB prbteins
(1) Bind to 'Origin of Replication site'
(2) Bind to single-stranded DNA only
(3) Bind to single stranded as well as dsDNA
(4) Bind to double stranded RNA

105. All of the following are found in the structure of deoxyadenosine, **except**
(1) N-glycosidic linkage
(2) Phosphate group
(3) Pentose sugar
(4) Purine

106. Choose the **wrong** match

(1)	F. Meischer	–	First identified DNA in nucleus
(2)	Wilkins & Franklin	–	X-ray diffraction data of DNA
(3)	Altman	–	Termed nuclein to DNA
(4)	E. Chargaff	–	Gave ratios between nitrogenous bases A and T and G and C for dsDNA

(1) (1) (2) (2)
(3) (3) (4) (4)

107. Following are given two statements, labelled as **Assertion (A)** and **Reason (R)**.
Assertion (A): We can predict the base sequence of the other strand of DNA if the sequence of one strand is known.
Reason (R): Both the strands of DNA are complementary to each other.
In the light of the above statements, choose the **correct** option.
(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**
(2) Both **(A)** and **(R)** are true but **(R)** is not the correct explanation of **(A)**
(3) **(A)** is true but **(R)** is false
(4) **(A)** is false but **(R)** is true

108. Transcription and replication share some similar features, as
(a) Both of them follow the rule of complementarity.
(b) Both of them produces a nucleotide chain that require splicing.
(c) Both of them utilises the main polymerising enzyme that perform the synthesis only in 5' → 3' direction.
(d) Both of them uses both strands of DNA as template
(e) Both of them occurs in the nucleus in eukaryotic cell.
Select the **correct** ones.
(1) (a), (c) and (e) only
(2) (b) and (d) only
(3) (b), (d) and (e) only
(4) (c), (d) and (e) only

109. In an organism following stretch of a polypeptide is synthesized.
Histidine – Cysteine – Methionine – Tryptophan – Aspartic acid
Codons for the given amino acids are as follows:
His – CAU or CAC
Cys – UGU or UGC
Met – AUG
Trp – UGG
Asp – GAU or GAC
Here in the options only one strand of the double stranded DNA has been represented. Then which among the following can be sequence of the template in DNA that codes for the given peptide?
(1) 3' GTCCCACATGCAGTG 5'
(2) 5' GTCCCATACGCAGTG 3'
(3) 5' GTCCCACATGCAGTG 3'
(4) 3' GTCCATCCGGCAGTG 5'

110. Read the following Assertion (A) and Reason (R) statements and select the **correct** option.
Assertion (A) : In *E. coli*, if a piece of DNA is needed to be propagated during recombinant DNA procedures, it requires a vector.
Reason (R) : There is a definite region in *E. coli* DNA, from where the replication originates and vector provides the additional origin of replication.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) (A) is false but (R) is true

111. Select the **incorrect** statement from the following.
(1) In most of the eukaryotes, the structural gene in monocistronic
(2) Exons are the expressed sequence of genes
(3) Introns do not appear in processed RNA
(4) A gene does not code for tRNA

112. Select the **incorrect** statement w.r.t. 'The adapter molecule'.
(1) Their existence was postulated by Francis Crick
(2) Was also known as soluble RNA (sRNA)
(3) Has a guanine residue at its 3'-end
(4) Also acts as intermediate molecule between triplet code of mRNA and amino acid sequence of polypeptide chain

113. Which one of the following is not a part of a transcription unit in DNA?
(1) A promoter
(2) A terminator
(3) The inducer
(4) The structural gene

114. Read the following statements and select the correct option.
Statement A: The discovery of nuclein by Meischer and the proposition for principles of inheritance by Mendel were almost at the same time.
Statement B: By 1926, the quest to determine the mechanism for genetic inheritance had reached the molecular level.
(1) Both statements are correct
(2) Both statements are incorrect
(3) Only statement A is correct
(4) Only statement B is correct

115. The role of a ribozyme (23S rRNA) during translation is to
(1) Break down the proteins into smaller fragments
(2) Synthesize mRNA
(3) Catalyse the formation of peptide bond
(4) Catalyse aminoacylation of tRNA

116. Observe the following sequence in DNA.
Coding strand → 5'– GACCCAAGCTACGG–3'
Which of the following option represents the **correct** sequence of RNA after transcription?
(1) 5' – CUGGGCCAUCGUGC – 3'
(2) 5' – CUGGGUUCGAUGCC – 3'
(3) 3' – GACCCAAGCTACGG – 5'
(4) 3' – GGCAUCGAACCCAG – 5'

117. Select the **correct** features of human genome from the following:
(a) It contains 3774.1 million base pairs.
(b) The average gene consists of 30000 bases and shows slight variation from gene to gene.
(c) Snips occur at about 1.4 million locations.
(d) Repetitive sequences show direct coding functions and shed light on the structure of a chromosome.
The correct one(s) is/are -
(1) (a) only
(2) (a) and (b) only
(3) (b), (c) and (d) only
(4) (c) only

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The last step of DNA fingerprinting is
(1) Blotting (2) Hybridisation
(3) Electrophoresis (4) Autoradiography

118. Operon model in bacteria was given by
(1) Watson and Crick (2) Jacob and Monod
(3) Severo Ochoa (4) Nirenberg

120. All of the following are the properties of genetic code, **except**
(1) Being triplet
(2) Nearly universal
(3) Read in contiguous fashion
(4) Being overlapping

121. Some amino acids are coded by more than one codon. It reflects which of the given feature of genetic code?
(1) Commaless punctuation (2) Degeneracy of code
(3) Non-overlapping (4) Universality of codes

122. Which of the following is **incorrect** w.r.t. levels of regulation of gene expression in eukaryotes?
(1) Replication level (2) Transcriptional level
(3) Processing level (4) Translation level

123. If adenine in the genetic material of $\phi \times 174$ bacteriophage is 40% of the total nitrogenous bases, then calculate the proportion of thymine in it, according to Chargaff's rule.
(1) 40%
(2) 70%
(3) 50%
(4) Cannot be determined by the rule

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124. How many amino acids will be coded by the sequence if the 12th base of given mRNA converts to 'G'?
5'AUGGUGUGAUACACC3'
(1) Two (2) Three
(3) Five (4) Four

125. Heterochromatin differs from euchromatin as the former are
(1) Active regions during transcription process
(2) Lightly stained regions
(3) Densely packed chromatin
(4) Absent in all eukaryotes

126. The existence of complementarity with DNA is not seen during the process of
(1) Replication (2) Reverse transcription
(3) Translation (4) Transcription

127. Select the correct statement(s) about *lac* operon.
a. Glucose or galactose act as inducer for the expression of *lac* operon
b. In the presence of lactose the repressor protein binds with the lactose molecules
c. It is an example of repressible operon system
d. Its regulation by repressor protein is called positive regulation.
(1) (a) and (b) (2) (b) and (c)
(3) Only (b) (4) Only (d)

128. Select the incorrect statement regarding the salient features of the double-helix structure of DNA
(1) It is made of two polynucleotide chains
(2) The two chains have anti-parallel polarity
(3) Guanine is bonded with thymine by three H-bonds
(4) The two chains are coiled in a right-handed fashion

129. What is incorrect for human chromosome 1?
(1) It is one of the largest chromosome
(2) Its sequence was completed in May 2007
(3) It has maximum number of genes
(4) It was the last chromosome to be sequenced

130. The non-human model organisms sequenced in Human Genome project were
(1) A nematode and fruit fly (2) Wheat and rice
(3) Fish and birds (4) Garden pea and fruit fly

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131. Activation of amino acids for the process of translation is done by
(1) Peptidyl transferase
(2) Aminoacyl-tRNA synthetase
(3) Methionine
(4) Initiation factors

132. The accessibility of promoter regions of bacterial DNA in many cases is regulated by the interaction of proteins with sequences termed
(1) Regulators (2) Structural genes
(3) Inhibitor genes (4) Operators

133. Movement of ribosome on mRNA is called
(1) Transcription (2) Translation
(3) Translocation (4) Protein synthesis

134. Amongst the following, which one has least number of nucleotides in its genome?
(1) $\phi \times 174$ bacteriophage (2) *E. coli*
(3) Human (4) Lambda phage

135. Poly A tail is present in
(1) mRNA of bacteria (2) tRNA of eukaryotes
(3) Promotor of bacteria (4) mRNA of eukaryotes

ZOOLOGY

136. *Homo sapiens* evolved in
(1) Asia (2) Europe
(3) America (4) Africa

137. Flippers of Penguins and Dolphins are examples of
(1) Convergent evolution (2) Industrial melanism
(3) Natural selection (4) Adaptive radiation

138. Complete the analogy by selecting the correct option.
Bobcat : Tasmanian tigercat :: Lemur : _____
(1) Numbat (2) Mole
(3) Spotted cuscus (4) Tasmanian wolf

138. Select the **incorrect** statement w.r.t. evolution.
(1) Life probably appeared 500 million years after the formation of Earth.
(2) The first cellular form of life appeared on Earth about 2000 mya.
(3) It is presumed that around 200 mya *Ichthyosaurs* evolved from amphibians.
(4) About 65 mya, the dinosaurs suddenly disappeared from Earth.

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140. Which of the following is **not** true statement?
(1) Evolution is a stochastic process
(2) Evolution is not a directed process in the sense of determinism
(3) Evolution is always from simple form to complex form
(4) Natural selection in which more individual acquire value other than the mean character value is called directional selection

141. Choose the oldest ancestor in evolution of man among the following.
(1) *Homo habilis* (2) *Ramapithecus*
(3) Neanderthal man (4) *Homo sapiens*

142. Choose the **correct** sequence of appearance of different plant forms originated during evolution.
(1) Chlorophyte ancestor → Tracheophyte ancestors → Rhynia type plants → Bryophytes
(2) Progymnosperm → Seed ferns → Monocotyledons → Dicotyledons
(3) Tracheophyte ancestors → *Rhynia* type plants → Psilophyton → Gnetales
(4) Psilophyton → Progymnosperm → Seed ferns → Conifers

143. The Neanderthal man had a cranial capacity of
(1) 900cc (2) 1400cc
(3) 650cc (4) 800cc

144. Choose the option which represents stabilising natural selection in which more individuals acquire mean character value.
(1) After industrialisation, high frequency of melanic moths, than white moths
(2) High frequency of white coloured moths before industrialisation
(3) Average body weight at which new born baby is more likely to survive
(4) Selection of large sized beak of birds during drought than small sized beak

145. Choose the **incorrect** match.
(1) Unit of evolution – Population
(2) Unit of natural selection – Individual
(3) Origin of Progymnosperms – Paleozoic era
(4) *Homo habilis* – Fossils discovered in Java in 1890

146. Read the following statements.
Statement A : During the course of evolution the pouched mammals of Australia survived.
Statement B : Placental mammals failed to be a competitor to marsupials as they were separated due to continental drift.
Choose the **correct** option.
(1) Only statement A is correct
(2) Both statements are correct
(3) Both statements are incorrect
(4) Only statement B is correct

147. "Excess use of herbicides has resulted in increased number of herbicide resistant varieties in a lesser time."
Which of the following is mainly responsible for the above given scenario?
(1) Anthropogenic action and natural selection
(2) Genetic drift and gene migration
(3) Convergent evolution and founder's effect
(4) Founder's effect and gene recombination

148. Choose the **incorrect** statement.
(1) Work by Thomas Malthus on populations influenced Darwin
(2) Lamarck suggested that evolution is driven by use and disuse of organs
(3) Mendel talked about inheritable 'factors' which influenced phenotypes of organisms
(4) Ernst von Baer suggested that evolution is based on reproductive fitness of individuals

149. Sweet potato and potato are examples of analogous structure as
(a) Former is root modification
(b) Latter is stem modification
(c) Former is stem modification
(d) Latter is root modification
Choose the **correct** option.
(1) (a) and (d) only (2) (a) and (b) only
(3) (b) and (c) only (4) (c) and (d) only

150. **Assertion (A):** Microbes have more chances to evolve faster than the fish or fowl.
Reason (R): The rate of appearance of new forms is linked to the life cycle or the life span.
In the light of above statements, choose the **correct** option.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

151. Read the following statements:
Statement A: All the existing life forms share similarities and share common ancestors.
Statement B: The geological history of Earth closely correlates with the biological history of Earth.
Choose the **correct** option.
(1) Both statements A and B are correct
(2) Both statements A and B are incorrect
(3) Only statement A is correct
(4) Only statement B is correct

152. The idea of 'Panspermia' states that
(1) Life appeared on Earth 4000 years back
(2) Units of life called spores were transferred to different planets from Earth
(3) Units of life were transferred to different planets excluding Earth
(4) Units of life were transferred to different planets including Earth

153. Select the option that represents incorrect match.
(1) Early Greek thinkers – Idea of Panspermia
(2) Louis Pasteur – Life comes only from pre-existing life
(3) Oparin – Disproved spontaneous generation theory
(4) Haldane – First form of life could have come from pre-existing non-living organic molecules

154. The essence of Darwinian theory of evolution is
(1) Use and disuse of organs
(2) Natural selection
(3) Genetic drift
(4) Gene recombination

155. In Hardy-Weinberg equilibrium, the frequency of heterozygous individual is represented by
(1) p^2 (2) 2 pq
(3) q^2 (4) pq

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156. One of the important consequences of geographical isolation is
(1) Inhibition of recombination during gametogenesis
(2) No change in the isolated fauna
(3) Increase in random mating
(4) Speciation through reproductive isolation

157. Presence of which phenomenon does not cause any change in genetic equilibrium of population?
(1) Mutation
(2) Gene flow
(3) Natural selection
(4) Random mating

158. Select the correct sequence of the division of geological time scale.
(1) Eras – Epochs – Periods
(2) Eras – Periods – Epochs
(3) Epochs – Eras – Periods
(4) Periods – Epochs – Eras

159. In the Urey Miller experiment, all of the following gases were taken in the closed flask **except**
(1) SO_2
(2) NH_3
(3) H_2
(4) CH_4

160. Choose the **incorrect** statement.
(1) Skull of adult modern man is similar to that of baby chimpanzee
(2) Modern *Homo sapiens* arose during ice age
(3) Cave paintings were discovered from Bhimbetka rock shelter in Andhra Pradesh
(4) Pre-historic cave art developed about 18000 years ago

161. A gene locus has two alleles A, a. If the frequency of recessive allele a is 0.8, then what will be the frequency of heterozygotes in a random mating population at equilibrium?
(1) 0.32
(2) 0.16
(3) 0.04
(4) 0.64

162. Read the following statements and choose the **correct** option.
Statement-A : *Tyrannosaurus* was a flesh eating dinosaur.
Statement-B : *Brachiosaurus* had big kite like plates on the back for its protection.
(1) Both statements are correct
(2) Both statements are incorrect
(3) Statement A is incorrect
(4) Statement B is incorrect

163. Two key postulates on which Lamarckism is based **include**
(a) Natural selection
(b) Inheritance of acquired characters
(c) Use and disuse of organs
(d) Branching descent
Choose the correct option.
(1) (a) & (d)
(2) (b) & (c)
(3) (c) & (d)
(4) (a) & (c)

164. How many of the plant forms mentioned in the box below did not originate from Psilophyton?

Zosterophyllum, Ginkgos, Conifers, Bryophytes, Ferns, Gnetales, Sphenopsids, Arborescent lycopods

Select the **correct** option.
(1) Three
(2) Five
(3) Four
(4) Two

165. Select the correct set of animals exhibiting homology.
(1) Mole and Marsupial mole
(2) Lemur and Spotted cuscus
(3) Flying squirrel and Bobcat
(4) Anteater and Numbat

166. Analogous structures are the result of
(1) Adaptive radiation
(2) Co-evolution
(3) Convergent evolution
(4) Divergent evolution

167. Who among the following implied natural selection as a mechanism of evolution?
(1) Lamarck
(2) Charles Darwin
(3) Hugo de Vries
(4) Ernst Heckel

168. Who finally dismissed the theory of spontaneous generation?
(1) S.L Miller
(2) A.I Oparin
(3) Louis Pasteur
(4) Ernst von Baer

169. Earth is supposed to have been formed about _____ years back. Choose the option which fills the blank **correctly**.
(1) 4.5 billion
(2) 4 million
(3) 20 billion
(4) 20 million

170. **Assertion (A) :** Industrial melanism illustrates the role of anthropogenic action in evolutionary process.
Reason (R) : Pollution generated by human activity led to origin of darker variety of moths.
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

171. The first organisms that appeared on Earth were probably
(1) Chemoheterotrophs
(2) Chemoautotrophs
(3) Photoautotrophs
(4) Decomposers

172. While explaining industrial melanism, agent of natural selection is
(1) Man
(2) Pollution
(3) Bird
(4) Moth

173. Reptilian ancestry of birds is shown by the presence of
(1) Scales on hind limbs
(2) Non-cleidoic eggs
(3) Teeth in beak of birds
(4) Three chambered heart

174. The presence of embryonic features such as tail and gill slits, in the embryo of all vertebrates supports the theory of
(1) Recapitulation
(2) Natural selection
(3) Retrogressive metamorphosis
(4) Germplasm

175. Among following, the most recent is
(1) *Homo sapiens*
(2) *Australopithecines*
(3) *Homo habilis*
(4) *Dryopithecus*

176. Fossils discovered in Java in 1891 belonged to
(1) *Homo habilis*
(2) Neanderthal man
(3) *Homo erectus*
(4) *Ramapithecus*

177. *Cycas* and *Ginkgo* are connecting links between
(1) Bryophytes and pteridophytes
(2) Pteridophytes and gymnosperms
(3) Gymnosperms and angiosperms
(4) Thallophyta and bryophyta

178. Mesozoic era is called the
(1) Age of Fishes
(2) Age of Reptiles
(3) Age of Mammals
(4) Age of Birds

179. Origin of life from non-living matter is known as
(1) Coacervates
(2) Abiogenesis
(3) Biogenesis
(4) Paedogenesis

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180. During the origin and evolution of life, key biological compounds were progressively synthesised in ocean with the help of energy obtained from
(1) Lightening only
(2) Ultraviolet light only
(3) Lightening as well as ultra-violet light
(4) Exclusively from combustion of certain compounds

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